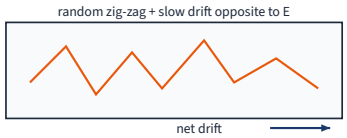
# CHAPTER 14 ELECTRIC CURRENT

**Key idea:** current is the drift of charge carriers; resistance depends on the material and its shape.

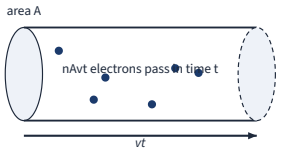
## 1 ELECTRIC CURRENT




$$I = \frac{dQ}{dt} \qquad Q = Ne$$

- Conventional current: direction of + charge flow
- Electrons drift the opposite way
- 1 A = 1 C s$^{-1}$

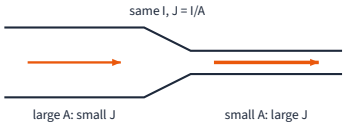
## 2 DRIFT VELOCITY




$$I = nAev_d$$

- $v_d \sim 10^{-4}$ m s$^{-1}$; random speed $\sim 10^5$–$10^6$ m s$^{-1}$
- Lamp lights at once: the **field** travels at nearly $c$
- Narrower wire: larger $v_d$

## 3 CURRENT DENSITY




$$J = \frac{I}{A} = nev_d \qquad J = \sigma E$$

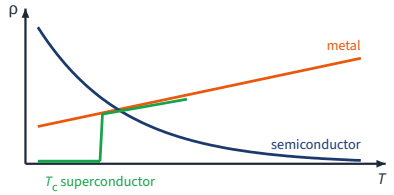
Vector, A m$^{-2}$; $J = \sigma E$ is the microscopic Ohm's law.

## 4 RESISTANCE & RESISTIVITY

$$R = \frac{V}{I} \qquad R = \frac{\rho L}{A} \qquad \sigma = \frac{1}{\rho}$$

- $\rho$ (Ω m) and $\sigma$ are properties of the **material**
- Stretched to $k$ × length (same volume): $R \times k^2$

## 5 EFFECT OF TEMPERATURE




- **Metal:** ions vibrate more, more collisions, so $\rho$ rises
- **Semiconductor:** more carriers released, so $\rho$ falls
- **Superconductor:** $\rho = 0$ below the critical temperature

## 6 MICROSCOPIC VIEW

$$\sigma = \frac{ne^2\tau}{m} \qquad v_d = \frac{eE\tau}{m}$$

$\tau$: mean time between collisions (relaxation time).

## 7 ELECTRICAL POWER

$$P = VI \qquad P = I^2R \qquad P = \frac{V^2}{R}$$

Energy $W = VIt$; 1 kW h = 3.6 × 10$^6$ J. Heater coil: combine $P = V^2/R$ with $R = \rho L/A$.

## 8 COMMON SECTION A TRAPS

- Using the diameter as radius in $A = \pi r^2$.
- Thinking drift speed is the speed of the signal.
- Same current, smaller area: $J$ and $v_d$ larger.
- Resistivity does not change when the wire is cut.

## EXAM TIPS

- ✓Write $I = nAev_d$ and rearrange before substituting.
- ✓Use ratios for wires of the same material.
- ✓Explain temperature effects with collisions and carriers.
- ✓Units: $n$ in m$^{-3}$, $A$ in m$^2$.